SS-S-111a
January 12, 1968
SUPERSEDING
Fed. Spec. SS-S-111
September 3, 1964

FEDERAL SPECIFICATION

SOUND CONTROLLING MATERIALS (TROWEL AND SPRAY APPLICATIONS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers acoustical materials for trowel or spray applications.

1.2 Classification.

1.2.1 Types. Acoustical material shall be of the following types and densities, as specified (see 6.2):

Type I - Acoustical cementitious materials.
Density A - 18 to 24 pounds per cubic foot.
Density B - Over 24 pounds per cubic foot.

Type II - Fibrous acoustical material.
Density C - 11 to 14 pounds per cubic foot.
Density D - Over 14 pounds per cubic foot.

1.2.2 Classes. Acoustical material shall be class 25, 75, 200, or over 200 in accordance with flame spread properties (see 3.4), and as specified (see 6.2).

1.2.3 Grades.

1.2.3.1 Acoustical materials shall be the specified grade (see 6.2) based on the required noise reduction coefficient (see 3.3.1).

1.2.3.2 For auditorium treatment, acoustical material shall be the specified grade (see 6.2) based on the sound absorption coefficients at 500 cycles per second (Hz) (see 3.3.2) in lieu of the grades in 1.2.3.1.

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2. APPLICABLE DOCUMENTS

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

SS-B-755 - Building Board, Asbestos-Cement; Flat.
UU-S-48 - Sacks, Shipping, Paper.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Standards:

C423 - Standard Method of Test for Sound Absorption of Acoustical Materials in Reverberation Rooms.
E84 - Method of Test for Surface Burning Characteristics of Building Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103.)

American Trucking Associations, Inc.

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P St., N.W., Washington, D.C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Certificate of compliance. Unless otherwise specified (see 6.2), acoustical material offered under this specification shall be certified by the contractor to have successfully completed the tests for sound absorption, flame spread index, and dust.

3.2 Material.

3.2.1 Type I. Type I ingredients shall be uniformly mixed, ready for use with the addition of water. When received, the material in

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the container shall be dry and shall not be caked or sticky. After it has been properly mixed with the amount of water specified by the manufacturer, the material shall be of such a consistency that it can be applied with a plastering trowel or with a suitable plaster-spraying machine. Upon drying, the acoustical material shall have a hardness adequate to form a durable surface.

3.2.2 Type II. Type II shall consist of mineral or vegetable fiber with an integrally mixed binder, or it may be supplied for use with a liquid binder added at the time of application.

3.3 Acoustical properties.

3.3.1 Noise-reduction coefficient (NRC). Acoustical materials used to reduce the noise level shall be classified in terms of sound-absorbing qualities at four frequencies: 250, 500, 1,000, and 2,000 Hz. The average of the sound-absorption coefficients at these four frequencies will be called the noise-reduction coefficient. For grading purposes, the average will be taken to the nearest multiple of 0.05. The value of the coefficient shall be determined in accordance with 4.4.1 and shall comply with the requirements of table I.

TABLE I. Grades for noise-reduction coefficient (NRC)

Grade	NRC
1	0.90 and over
2	0.80 -0.89
3	.70 - .79
4	.60 - .69
5	.50 - .59
6	.40 - .49
7	Less than 0.40

3.3.2 Sound absorption coefficient. For auditorium treatments, acoustical material specified in the invitation for bids shall be graded in terms of sound absorbing qualities at 500 Hz. The limiting values of sound absorption coefficients at 500 Hz for each grade shall be determined in accordance with 4.4.1 and shall comply with the requirements of table II.

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TABLE II. Grades for sound-absorption coefficients at 500 Hz (for auditorium treatment)

Grade	Coefficient at 500 Hz
101	0.90 and over
102	0.80 -0.89
103	.70 - .79
104	.60 - .69
105	.50 - .59
106	.40 - .49
107	Less than 0.40

3.4 Surface-flame spread. When tested in accordance with 4.4.2, acoustical material shall be designated as follows:

3.4.1 Class 25. The flame-spread index of the acoustical material shall not exceed 25 nor shall the material show evidence of continued progressive combustion. Materials with a flame-spread index of 25 or less are considered noncombustible.

3.4.2 Class 75. The flame-spread index shall not exceed 75 on the exposed surface. Materials with a flame-spread index of 75 or less are considered fire-resistant.

3.4.3 Class 200. The flame-spread index shall not exceed 200 on the exposed surface. Material having a flame-spread index of 200 or less is considered slow burning. Materials having a flame-spread index over 200 are considered combustible.

3.5 Dust test.

3.5.1 Civil agencies. Unless otherwise specified (see 6.2), the dust test shall not be required for civil agency procurement.

3.5.2 Military agencies. When subjected to the test specified in 4.4.3, dust removed from the acoustical material shall not exceed 0.01 gram per square foot.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Inspection lot. All acoustical material of the same type, density, class, and grade offered for delivery at one time shall be considered a lot for purposes of inspection.

4.3 Sampling for tests.

4.3.1 Sound absorption test and noise reduction test. Each sample for sound absorption and noise reduction tests shall consist of enough acoustical material, ready for application when mixed with water, to cover an area of approximately 72 square feet.

4.3.2 Flame-spread test. The sample for flame-spread test shall consist of enough material, ready for application with the possible exception of water only, to cover an area approximately 50 square feet.

4.3.3 Dust test. The sample for the dust test shall consist of enough material, ready for application with the exception of water, to cover an area of not less than 4 square feet.

4.4 Test methods.

4.4.1 Sound absorption and noise reduction coefficient tests. Sound absorption and noise reduction coefficients of the acoustical sample material shall be determined in accordance with ASTM C423.

4.4.2 Flame-spread index. The flame-spread index shall be determined in accordance with ASTM E84.

4.4.3 Dust test. The dust test sample shall be tested as specified in 4.4.3.2 and 4.4.3.4.

4.4.3.1 Alternate dust test. Upon approval of the contracting officer (see 6.2), an alternate dust test may be substituted for the test described in 4.4.3.2 and 4.4.3.3 provided the requirements of 3.5 have been met.

4.4.3.2 Specimen. The sample of acoustical material in the type and density specified shall be applied to an asbestos-cement sheet conforming to SS-B-755, type F, 3/8-inch thick. Application shall be in strict accordance with the manufacturer's printed instructions, including any prescribed surface preparation and finish treatment, except that not less than 0.5 inch of acoustical material shall be applied. Specimen shall be conditioned at 21° ± 1° C. (72° ± 2° F.) and 50 percent relative humidity until the weight of the specimen, weighed at 24-hour intervals, becomes constant to within 0.1 percent.

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4.4.3.3 Apparatus. The test apparatus shall consist of a tight metal duct approximately 17 inches wide by 24 inches high by 7 feet long. A centrifugal blower, driven by a 7.5 h.p. motor, shall be coupled to the duct. One side of the duct shall have an opening large enough to insert a test sample 18 inches wide by 48 inches long, having an exposed surface area to the air flow of approximately 4 square feet (see figure 1). Two dust collecting filters, with a 95 percent efficiency in filtering particles 5 microns in size and larger shall be installed upstream of the opening for the test sample (see 6.7). A black dust collecting cloth shall be placed downstream of the opening for the test sample. The cloth shall contain 76 warp ends per inch and 90 filler picks per inch and shall weigh 1.90 ± 0.1 ounces per square yard. The cloth shall be conditioned the same as the specimen (see 4.4.3.2).

4.4.3.4 Procedure. The test for dusting shall be conducted in a reasonably dust-free environment. A nondusting specimen shall be inserted in the test opening in the duct side wall. The complete system shall be purged for 24 hours at a minimum impinging velocity on the filter of 800 feet per minute and a minimum issuing velocity of 270 feet per minute. After purging for 24 hours the dust collecting cloth shall be removed and weighed to the nearest 0.0001 grams. The dust-collecting cloth shall be reinserted in the test apparatus. The acoustical sample shall be inserted in the test opening and subjected to the test for one hour. The dust collecting cloth shall then be again removed and reweighed. The net difference in weight between the dust collected with the nondusting specimen and dust collected with the test specimen shall constitute the dust removed from the acoustical material. The weight of dust removed shall not exceed the amount specified in 3.5.2.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping sacks fully prepared for delivery. The lot size shall be the number of shipping sacks in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.

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TABLE III. Classification of preparation for delivery defects (Continued)

Examine	Defects
Workmanship	Inadequate application of components such as incomplete closure of sacks.
Contents	Amount per container is more or less than required.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The material shall be furnished in sacks in accordance with the level A requirements of UU-S-48 for building plaster as indicated for sack construction number applicable to the weight dimensions involved.

5.1.2 Level B. The material shall be furnished in sacks in accordance with the level B requirements of UU-S-48 for building plaster as indicated for sack construction number applicable to the weight dimensions involved.

5.1.3 Level C. The materials shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.2 Marking (see 6.2).

5.2.1 Civil agencies. In addition to any special marking specified in the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special markings specified in the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Acoustical material is intended for use in the control of reverberation in auditoriums or other rooms. Type I

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material is normally applied over gypsum and portland cement base coats, and over metal laths and clip-systems. Type II material is intended for spraying beams, roof decks, and columns.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type and density of acoustical material required (see 1.2.1, 1.2.3.1, and 1.2.3.2).

(c) Class of material required (see 1.2.2, 3.4, 4.4.2, and 6.3).

(d) Grade of acoustical material required (see 1.2.3, 3.3.1, and 3.3.2).

(e) Certification of test data if different (see 3.1).

(f) If dust test is required by civil agencies (see 3.5.1).

(g) If alternate dust test is acceptable (see 3.5.2 and 4.4.3.1).

(h) Selection of applicable level of packing required (see 5.1).

(i) Marking required (see 5.2).

6.3 Classification change. Acoustical material having fire-resistant properties, previously classified as classes A, B, C, and D, in SS-A-111, are comparable to the following flame-spread classes:

Class A - Class 25
Class B - Class 75
Class C - Class 200
Class D - Over 200

6.4 Material offered by the contractor should be similar in all respects to material tested.

6.5 Paintability. When acoustical material covered by this specification is to be painted, a nonbridging paint should be used to prevent damage to the sound-absorption property.

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6.6 To insure proper and uniform thickness it is recommended that acoustical materials be applied between spot ground of proper size, spaced at intervals of about eight feet, or by other means recommended by the manufacturer to give equavalent results. After application the acoustical material should be finished with a cork float, steel trowel, roller, or other means to a true surface.

6.7 Filters. The Farr HP-2A filter, used as part of the test apparatus, is manufactured by the Farr Company, P. O. Box 90187-T, Airport Station, Los Angeles, California and is suitable for filtering particles as small as 5 microns.

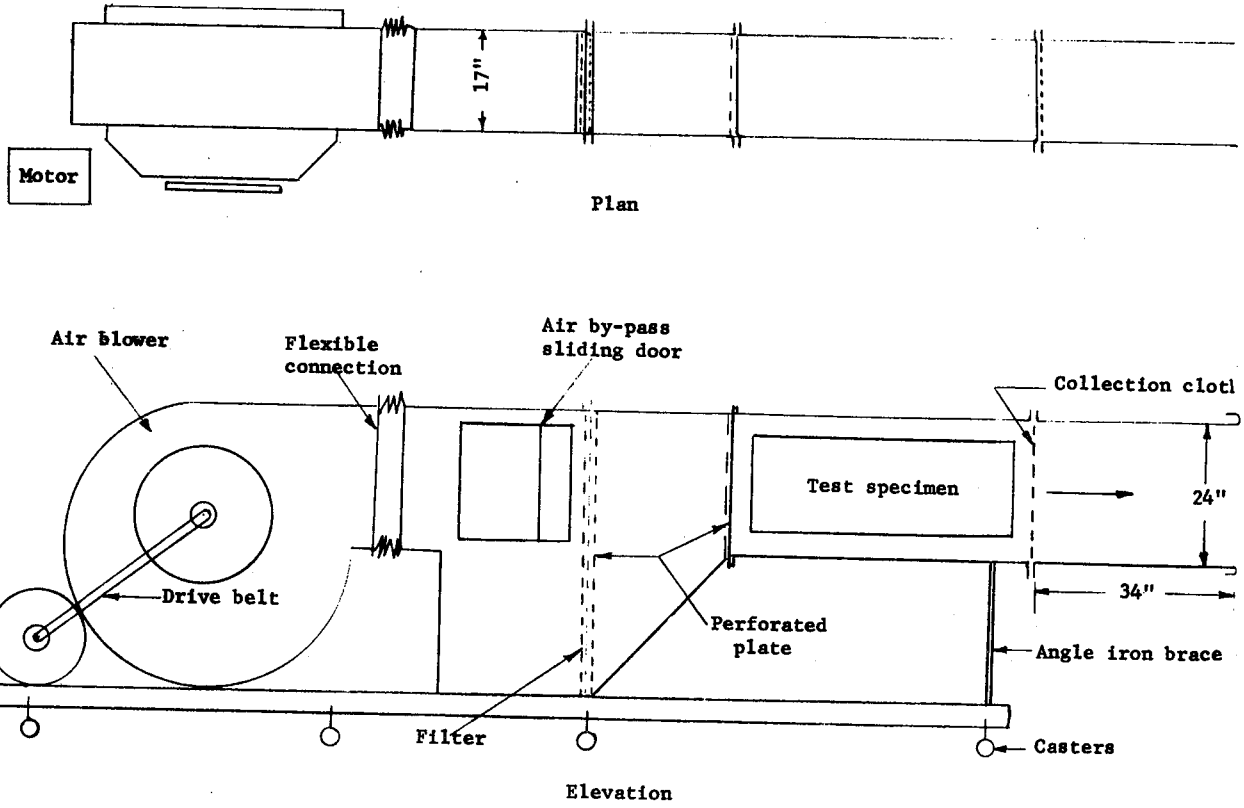


Figure 1. Typical apparatus for dust test

MILITARY CUSTODIANS:
Army - ME
Navy - YD
Air Force -884

Review activities:
Army - CE
Navy - YD
Air Force - 84

User activity:
Army - ME

Preparing activity:
Navy - YD

CIVIL AGENCIES INTEREST:
GSA-FSS
VA
COM-NBS

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